Overview & Design

Software Development: Hospital Project

Overview & Design

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Overview

The primary goal is to create a robust, object-oriented software solution that manages the lifecycle of medical data. The system is designed to handle different types of users. Such as **Doctors**, **Admins**, and **Insurance Admins.** to ensure that data is recorded and retrieved accurately while maintaining a high standard of programming practice.

- **Platform**: Java (Object-Oriented Paradigm).
- **Database**: MariaDB (Hospital database).
- **Core Logic**: Implementation of inheritance, encapsulation (private variables), and polymorphism.
- **Key Deliverables**: UML diagrams (Class and Use Case), source code, a testing plan, a user guide, and an evaluation report.

Functional Requirements

The functional requirements define what the system must do to support the business needs of the health insurance company.

1. Patient & Insurance Management

a. **Register Patients**: The system must store patient names, addresses, phone numbers, and emails.
b. **Track Insurance**: Each patient must be linked to a primary care doctor and an insurance company via an insurance ID number.
c. **Verify Eligibility**: The system must allow an Insurance Admin to verify patient coverage and manage provider lists.

2. Clinical Operations (Doctor Tasks)

a. **Record Visits**: Doctors must be able to log patient visits, including the date of the visit, symptoms reported, and the diagnosis given.
b. **Issue Prescriptions**: The system must track prescribed drugs, including the drug name, dosage, duration, and purpose.
c. **Safety Checks**: The application should provide information on a drug's possible side effects to ensure there are no conflicts with a patient's other medications.

3. Data Retrieval & Reporting

a. **Medical History**: Users must be able to retrieve historical visit and prescription data.
b. **Search Filters**: The system should allow for filtered searches, such as viewing all prescriptions for a specific patientID or viewing the most recent entries.
c. **Trend Tracking**: The accumulated data will eventually be used to track health trends and perform modeling.

4. System Administration

a. **Staff Management**: Admins must be able to manage doctor profiles, including their specialties and contact information.
b. **Database Maintenance**: Admins are responsible for updating the drug database and ensuring system integrity.

Non-Functional Requirements

These define the operational constraints and quality attributes of the system. Not all will be part of the system and are things that will be implemented in the future.

- **Security**: All sensitive medical data must be encrypted at rest and in transit. Access is strictly role-based, ensuring Doctors cannot perform Admin tasks and vice versa.
- **Data Integrity**: The MariaDB backend must use ACID-compliant transactions to ensure that patient records are never partially saved or corrupted during a crash.
- **Performance**: The Dynamic Query Builder must return search results in under 2 seconds to maintain efficiency in a fast-paced medical environment.
- **Usability**: The interface must be intuitive enough for medical staff to record a visit with minimal typing.
- **Scalability**: The system architecture should allow for an increasing number of patient records and concurrent users without a significant dip in response time.

Task & User Analysis

This section identifies who is using the system and the core tasks they must accomplish.

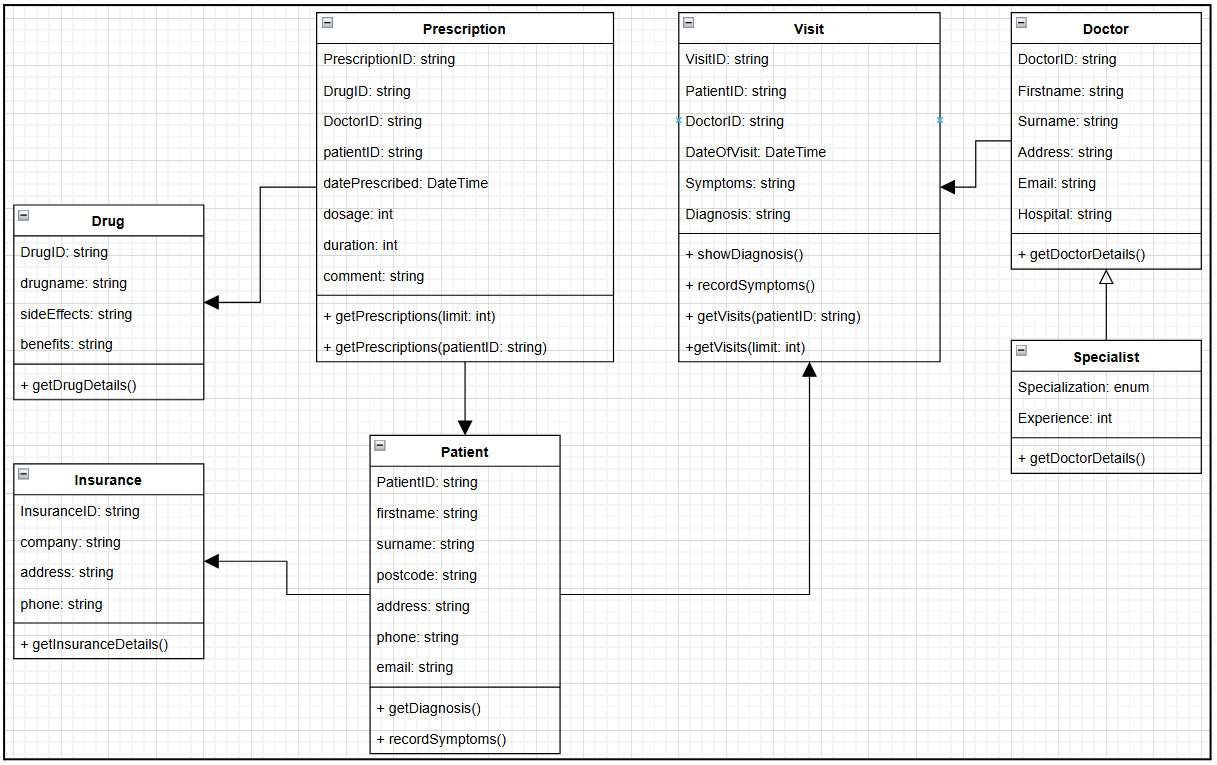
Persona	Goal	Technical Skill
Doctor	Quickly log patient symptoms and safely prescribe medication.	Moderate; requires a streamlined workflow.
Insurance Admin	Verify patient eligibility and manage insurance provider details.	High; focused on data accuracy and compliance.
System Admin	Maintain the drug database and manage staff access levels.	Expert; requires full system visibility.

Key Task Flows

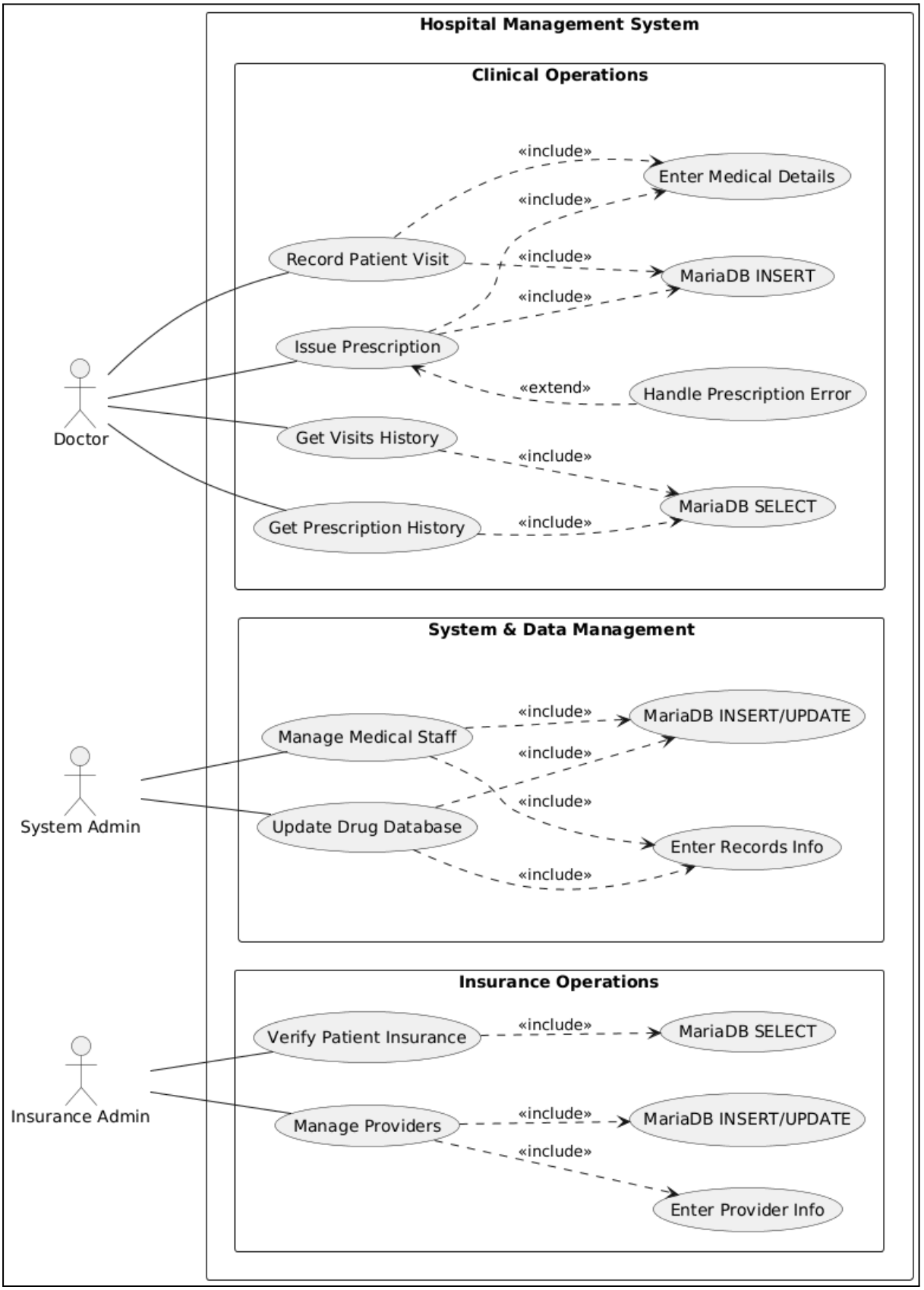
1. Patient Verification (Doctor/Admin):
    - Goal: Access a patient record.
    - Action: Input "Rule of Three" identifiers (First Name, Surname, Address) to unlock the medical history.
2. Clinical Documentation (Doctor):
    - Goal: Record a visit.
    - Action: Select a patient, input symptoms, and use the safety check feature to ensure a new prescription doesn't conflict with existing meds.
3. Database Maintenance (System Admin):
    - Goal: Update the drug repository.
    - Action: Upload new drug profiles, including dosage guidelines and known side effects.

Designs

Class Diagram

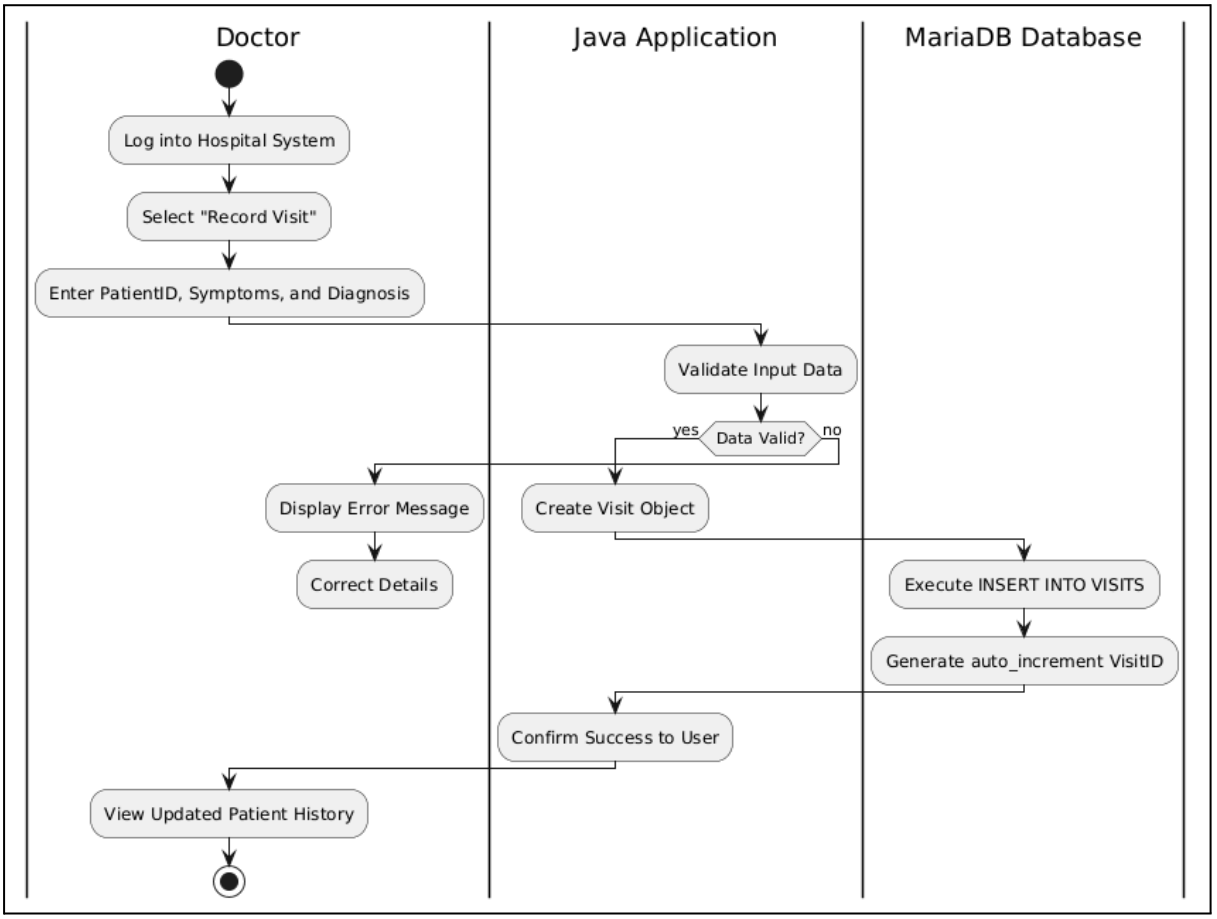


Use Case Diagram

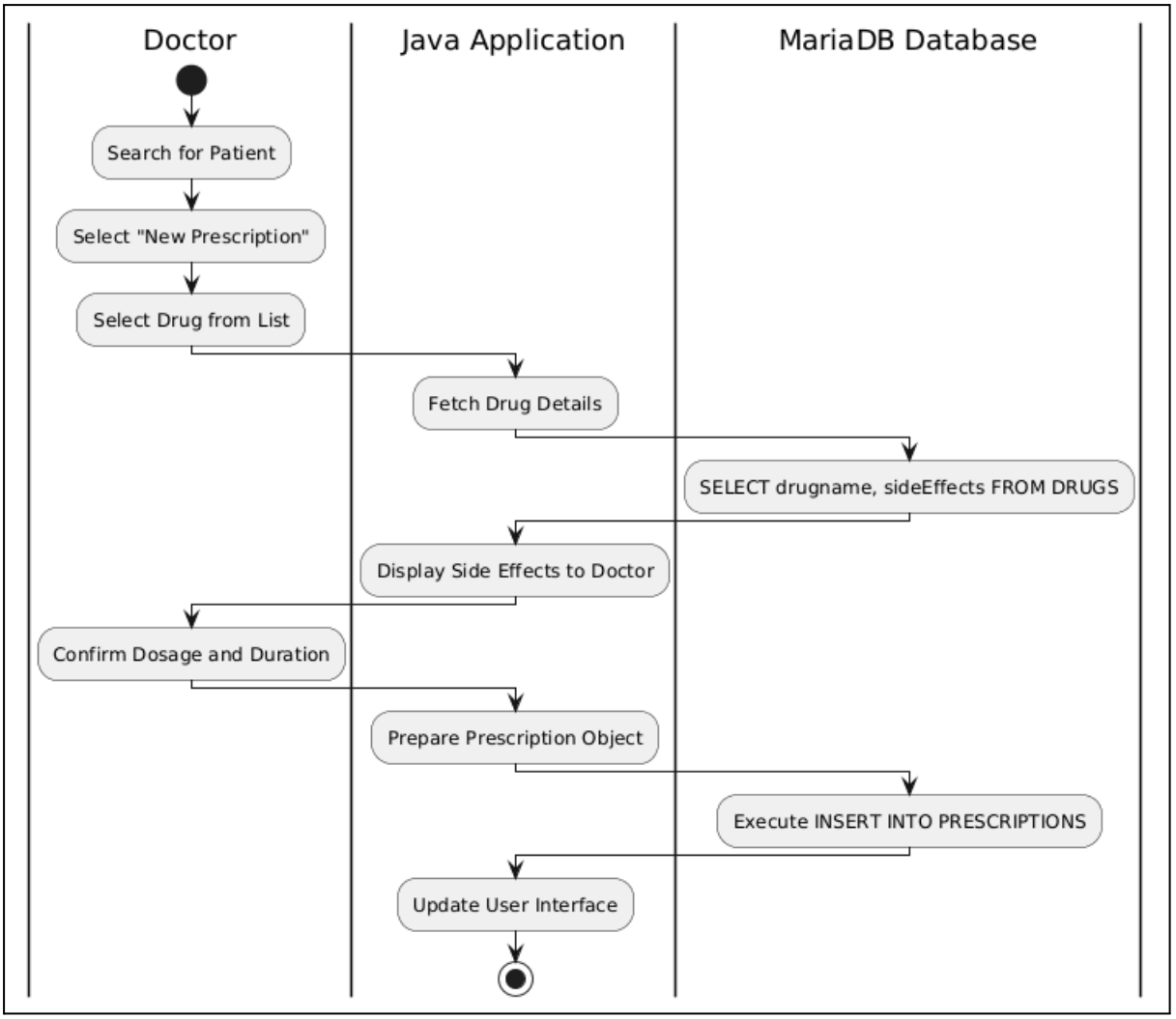


Activity Diagram

Recording Visits



Issuing a new prescription



Phase 1: Some Thoughts

I'm thinking of integrating a **Dynamic Query Builder** that would allow staff to find patients using real world identifiers (like name and address) rather than a hard to remember ID. To ensure **GDPR compliance**, the search requires multiple points of verification before sensitive medical data is revealed, ensuring that private information is only accessed after the patient's identity is confirmed. By using real world standards I will implement the **rule of two** before anyone can access data. For example the required inputs are: Firstname, Surname. Optional inputs can be added to refine searches and narrow down results.

User Guide

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User Guide

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Getting the Files

To get the required files like main.java and the database you will first need to get access to the github and download the folder.
https://github.com/Jayden-Wilson-01/Hospital-System

Database

The database has already been created and is ready to use however you will need to change the Database.java file to contain the correct url details like username and password.

Navigation

The application uses a numeric selection system. To perform an action: Type the number corresponding to your desired task, then press Enter to confirm

Navigating Main Menu

Module	Selection
Patients	1
Doctor	2
Prescriptions	3
Drugs	4
Visits	5
Insurance	6
Conflicts	7

Navigating Patients Menu

Module	Selection
Exit	0
Add patient	1
Update patient	2
Delete patient	3
View all patients	4
View patient by ID	5
View patient by parameters	6

Navigating Doctor Menu

Module	Selection
Exit	0
Add doctor	1
Update doctor	2
Delete doctor	3
View all doctors	4
View doctor by ID	5
View doctor by parameters	6
View doctor by speciality	7

Navigating Prescription Menu

Module	Selection
Exit	0
Add prescription	1
Update prescription	2
Delete prescription	3
View all prescriptions	4
View prescription by ID	5
View prescription by parameters	6

Navigating Drugs Menu

Module	Selection
Exit	0
Add drug	1
Update drug	2
Delete drug	3
View all drugs	4
View drug by ID	5
View drug by parameters	6

Navigating Visits Menu

Module	Selection
Exit	0
Add visit	1
Update visit	2
Delete visit	3
View all visit	4
View visit by ID	5
View visit by parameters	6

Navigating Insurance Menu

Module	Selection
Exit	0
Add insurance	1
Update insurance	2
Delete insurance	3
View all insurance	4
View insurance by ID	5
View insurance by parameters	6

Navigating Conflicts Menu

Module	Selection
Exit	0
Add conflict	1
Delete conflict	2
View conflict by drug ID	3

Manage Patients

1. To **create** a patient type 1 then enter the data for the fields that show. It will ask if you want to include **insurance** and **primary doctor**, these are **optional**.

```
Enter an option (0-6): 1
Enter Patient ID or 0 to cancel: patient600
Include insurance? [yes] [no]: yes
Enter Insurance ID or 0 to cancel: MJ2JSHG
Include primary doctor? [yes] [no]: yes
Enter Doctor ID or 0 to cancel: 3527
Enter patients firstname or 0 to cancel: jayden
Enter patients surname or 0 to cancel: wilson
Enter patients address or 0 to cancel: 10 westpark mintlaw, AB425HY
Enter patients phone or 0 to cancel: 07368350709
Enter patients email or 0 to cancel: jayden@gmail.com
```

2. To **update** a patient type 2 then enter the data for the fields that show. It will ask if you want to include **insurance** and **primary doctor**, these are **optional**.

```
Enter an option (0-6): 2
Enter Patient ID or 0 to cancel: patient600
Include insurance? [yes] [no]: no
Include primary doctor? [yes] [no]: yes
Enter Doctor ID or 0 to cancel: 3527
Enter patients firstname or 0 to cancel: jayden-james
Enter patients surname or 0 to cancel: wilson
Enter patients address or 0 to cancel: address
Enter patients phone or 0 to cancel: phone number
Enter patients email or 0 to cancel: email
```

3. To **delete** a patient type 3 then enter the data for the fields that show.

```
Enter an option (0-6): 3
Enter Patient ID or 0 to cancel: patient600
```

4. To **view all** patients type 4 then enter the data for the fields that show. The limit sets a hard limit on how many patients to show.

```
Enter limit: 2

Patient ID: 01YB9G9E
Insurance Info:
    ID: MJ2JSHG
    Name: Mary Kay Inc
Primary Doctor Info:
    ID: null
    Name: null null
Patient firstname: Iain
Patient surname: Ferrant
Patient address: EH3 9UJ 026 Larry Drive
Patient phone: 747 852 2741
Patient email: iferrantg@amazon.de
```

5. To **view** patients **by id** type 5 then enter an existing patient id.

```
Enter Patient ID or 0 to cancel: PHXNAQJ0

Patient ID: PHXNAQJ0
Insurance Info:
    ID: NHS001
    Name: NHS
Primary Doctor Info:
    ID: null
    Name: null null
Patient firstname: Janelle
Patient surname: Behnecken
Patient address: EH2 8BC 26 Dunning Parkway
Patient phone: 703 686 0183
Patient email: jbehneckene@craigslist.org
```

6. To **view** patients **by firstname and surname** type 6 then enter the data for the fields that show

```
Patient ID: PHXNAQJ0
Insurance Info:
    ID: NHS001
    Name: NHS
Primary Doctor Info:
    ID: null
    Name: null null
Patient firstname: Janelle
Patient surname: Behnecken
Patient address: EH2 8BC 26 Dunning Parkway
Patient phone: 703 686 0183
Patient email: jbehneckene@craigslist.org
```

Manage Doctors

7. To **create** a doctor type 1 then enter the data for the fields that show. It will ask if the doctor is a **specialist**, which is **optional**.

```
Is the doctor a specialist? [yes] [no]: no
Enter Doctor ID or 0 to cancel: doctor400
Enter doctors firstname or 0 to cancel: jenny
Enter doctors surname or 0 to cancel: smith
Enter doctors address or 0 to cancel: 10 westpakr mintlaw ab425hy
Enter doctors email or 0 to cancel: jenny@outlook.com
Enter doctors hospital or 0 to cancel: mintlaw
```

8. To **update** a doctor type 2 then enter the data for the fields that show. It will ask if the doctor is a **specialist**, which is **optional.**

```
Is the doctor a specialist? [yes] [no]: no
Enter Doctor ID or 0 to cancel: doctor400
Enter doctors firstname or 0 to cancel: jenny anne
Enter doctors surname or 0 to cancel: smith
Enter doctors address or 0 to cancel: 10 westpark mintlaw
Enter doctors email or 0 to cancel: jenny@outlook.com
Enter doctors hospital or 0 to cancel: mintlaw
```

9. To **delete** a doctor type 3 then enter the data for the fields that show.

```
Enter an option (0-7): 3
Enter Doctor ID or 0 to cancel: doctor400
```

10. To **view all** doctors type 4 then enter the data for the fields that show. The limit sets a hard limit on how many doctors to show.

```
Enter a load limit: 2

Doctor ID: 1724
Doctor firstname: Boothe
Doctor surname: Pavlov
Doctor address: Suite 67
Doctor email: bpavlovd@liveinternet.ru
Situated hospital: null
Specialization: Ophthalmology
Experience: 0

Doctor ID: 3326
Doctor firstname: Appolonia
Doctor surname: Streatfeild
Doctor address: PO Box 77011
Doctor email: astreatfeild7@jugem.jp
Situated hospital: City Centre
Specialization: Oncologists
Experience: 0
```

11. To **view** doctor **by id** type 5 then enter the data for the fields that show.

```
Enter Doctor ID or 0 to cancel: 3326

Doctor ID: 3326
Doctor firstname: Appolonia
Doctor surname: Streatfeild
Doctor address: PO Box 77011
Doctor email: astreatfeild7@jugem.jp
Situated hospital: City Centre
Specialization: Oncologists
Experience: 0
```

12. To **view** doctors **by doctor firstname and surname** type 6 then enter the data for the fields that show.

```
Enter doctors firstname or 0 to cancel: jenny
Enter doctors surname or 0 to cancel: smith

Doctor ID: doctor400
Doctor firstname: jenny
Doctor surname: smith
Doctor address: 10 westpakr mintlaw ab425hy
Doctor email: jenny@outlook.com
Situated hospital: mintlaw
```

13. To **view** doctors **by speciality** type 7 then enter the data for the fields that show. .

```
Specialization Options
1: CARDIOLOGY
2: NEUROLOGY
3: PEDIATRICS
4: ONCOLOGY
5: ORTHOPEDICS
6: PSYCHIATRY
Enter doctors specialization (1-6) or 0 to cancel: 2

Doctor ID: doc10
Doctor firstname: jayden
Doctor surname: wilson
Doctor address: address
Doctor email: email
Situated hospital: hospital
Specialization: NEUROLOGY
Experience: 10
```

Manage Prescriptions

14. To **add** a prescription type 1 then enter the data for the fields that show.

```
Enter Prescription ID or 0 to cancel: prescription500
Enter Patient ID or 0 to cancel: 01YB969E
Enter Drug ID or 0 to cancel: 14060
Enter Doctor ID or 0 to cancel: 1724
Enter prescription date (yyyy-mm-dd) or 0 to exit: 2025-08-13
Enter dosage or 0 to exit: 5
Enter duration or 0 to exit: 5
Enter comment or 0 to exit: comment here
```

15. To **update** a prescription type 2 then enter the data for the fields that show.

```
Enter an option (0-6): 2
Enter Prescription ID or 0 to cancel: prescription100
Enter Patient ID or 0 to cancel: 01YB969E
Enter Drug ID or 0 to cancel: 1026382
Conflict found: BARIUM CATION (14060)
Enter Drug ID or 0 to cancel: 15493
Enter Doctor ID or 0 to cancel: 1724
Enter prescription date (yyyy-mm-dd) or 0 to exit: 2025-04-04
Enter dosage or 0 to exit: 5
Enter duration or 0 to exit: 5
Enter comment or 0 to exit: 5 pills a week for 5 weeks
```

16. To **delete** a prescription type 3 then enter the data for the fields that show.

```
Enter an option (0-6): 3
Enter Prescription ID or 0 to cancel: prescription100
```

17. To **view all** prescription type 4 then enter the data for the fields that show.The limit sets a hard limit on how many prescriptions to show.

```
Enter limit: 2

Prescription ID: 1383892245
Patient Info:
    ID: FZJXKBND
    Name: Skyler Chichgar
Doctor Info:
    ID: 9947
    Name: Ole Hadcock
Drug Info:
    ID: 34154
    Name: risperidone
Prescription date: 2023-04-08
Dosage: 16
Duration: 98
Comment: Animal-drawn vehicle accident injuring other specified person
```

18. To **view a** prescription **by id** type 5 then enter the data for the fields that show.

```
Enter Prescription ID or 0 to cancel: prescription100

Prescription ID: prescription100
Patient Info:
    ID: 01YB969E
    Name: Iain Ferrant
Doctor Info:
    ID: 1724
    Name: Boothe Pavlov
Drug Info:
    ID: 14060
    Name: BARIUM CATION
Prescription date: 2025-04-18
Dosage: 5
Duration: 5
Comment: test
```

19. To **view a** prescription **by patients name and doctors name** type 6 then enter the data for the fields that show.

```
Enter patients first name or 0 to cancel: Iain
Enter patients surname or 0 to cancel: Ferrant
Enter doctors first name or 0 to cancel: Boothe
Enter doctors surname or 0 to cancel: Pavlov

Prescription ID: prescription100
Patient Info:
    ID: 01YB9G9E
    Name: Iain Ferrant
Doctor Info:
    ID: 1724
    Name: Boothe Pavlov
Drug Info:
    ID: 14060
    Name: BARIUM CATION
Prescription date: 2025-04-18
Dosage: 5
Duration: 5
Comment: test

Prescription ID: prescription500
Patient Info:
    ID: 01YB9G9E
    Name: Iain Ferrant
Doctor Info:
    ID: 1724
    Name: Boothe Pavlov
Drug Info:
    ID: 14060
    Name: BARIUM CATION
Prescription date: 2025-08-13
Dosage: 5
Duration: 5
Comment: comment here
```

Manage Drugs

20. To **create** a drug type 1 then enter the data for the fields that show.

```
Enter Drug ID or 0 to cancel: drug315
Enter drug name or 0 to exit: moscovium
Enter side effects or 0 to cancel: death
Enter benefits or 0 to cancel: none to be found
```

21. To **update** a drug type 2 then enter the data for the fields that show.

```
Enter an option (0-6): 2
Enter Drug ID or 0 to cancel: drug315
Enter drug name or 0 to exit: moscovium
Enter side effects or 0 to cancel: death
Enter benefits or 0 to cancel: none
```

22. To **delete** a drug type 3 then enter the data for the fields that show.

```
Enter an option (0-6): 3
Enter Drug ID or 0 to cancel: drug315
```

23. To **view all** drugs type 4 then enter the data for the fields that show. The limit sets a hard limit on how many drugs to show.

```
Enter limit: 2

Drug id: 1026382
Drug name: test
Side effects: test
Benefits: test

Drug id: 10987654
Drug name: test
Side effects: test
Benefits: test
```

24. To **view** drug **by id** type 5 then enter the data for the fields that show.

```
Enter Drug ID or 0 to cancel: drug315

Drug id: drug315
Drug name: moscovium
Side effects: death
Benefits: none
```

25. To **view** drugs **by drug name** type 6 then enter the data for the fields that show.

```
Enter drugs name or 0 to cancel: moscovium

Drug id: drug315
Drug name: moscovium
Side effects: death
Benefits: none
```

Manage Insurance

26. To **create** an insurance type 1 then enter the data for the fields that show.

```
Enter Insurance ID or 0 to cancel: insurance145
Enter insurance company name or 0 to cancel: HealthCo
Enter insurance address or 0 to cancel: 42 north avenue abereen AB436HW
Enter insurance phone or 0 to cancel: 345678765
```

27. To **update** an insurance type 2 then enter the data for the fields that show.

```
Enter Insurance ID or 0 to cancel: insurance145
Enter insurance company name or 0 to cancel: HealthCo
Enter insurance address or 0 to cancel:  42 north avenue aberdeen AB436HW
Enter insurance phone or 0 to cancel: 345678765
```

28. To **delete** an insurance type 3 then enter the data for the fields that show.

```
Enter an option (0-6): 3
Enter Insurance ID or 0 to cancel: insurance145
```

29. To **view all** insurance type 4 then enter the data for the fields that show. The limit sets a hard limit on how many insurance companies to show.

```
Enter load limit: 3

Insurance ID: 54ARPD9
Company name: Cantrell Drug Company
Address: 17668 Nova Pass
Phone: 964-658-4951

Insurance ID: 9BFYJXV
Company name: Carilion Materials Management
Address: 068 Hollow Ridge Parkway
Phone: 478-698-2522

Insurance ID: BS03EM0
Company name: Bay State Medical
Address: 61735 Northfield Circle
Phone: 212-489-0603
```

30. To **view** insurance **by id** type 5 then enter the data for the fields that show.

```
Enter Insurance ID or 0 to cancel: insurance145

Insurance ID: insurance145
Company name: HealthCo
Address:  42 north avenue aberdeen AB436HW
Phone: 345678765
```

31. To **view** drugs **by company name** type 6 then enter the data for the fields that show.

```
Enter insurance company name or 0 to cancel: healthCo

Insurance ID: insurance145
Company name: HealthCo
Address:  42 north avenue aberdeen AB436HW
Phone: 345678765
```

Manage Visits

32. To **create** a visit type 1 then enter the data for the fields that show.

```
Enter an option (0-6): 1
Enter Visit ID or 0 to cancel: visit124
Enter Doctor ID or 0 to cancel: 9947
Enter Patient ID or 0 to cancel: FZJXKBND
Enter date of visit (yyyy-mm-dd) or 0 to exit: 2023-04-08
Enter symptoms or 0 to exit: headache
Enter diagnosis or 0 to exit: Animal drawn vehicle accident injuring other specified person
```

33. To **update** a visit type 2 then enter the data for the fields that show.

```
Enter Visit ID or 0 to cancel: visit124
Enter Doctor ID or 0 to cancel: 9947
Enter Patient ID or 0 to cancel: FZJXKBND
Enter date of visit (yyyy-mm-dd) or 0 to exit: 2023-08-08
Enter symptoms or 0 to exit: headache
Enter diagnosis or 0 to exit: Animal drawn vehicle accident injuring other specified person
```

34. To **delete** a visit type 3 then enter the data for the fields that show.

```
Enter an option (0-6): 3
Enter Visit ID or 0 to cancel: visit124
```

35. To **view all** visits type 4 then enter the data for the fields that show. The limit sets a hard limit on how many visits to show.

```
Enter limit: 2

VisitID: 1
Patient Info: ID: PHXNAQJ0 Name: Janelle Behnecken
Doctor Info: ID: 9947 Name: Ole Hadcock
DateOfVisit: 2023-05-06
Symptoms: Family history of other specified malignant neoplasm
Diagnosis: 168

VisitID: 10
Patient Info: ID: FZJXKBND Name: Skyler Chichgar
Doctor Info: ID: 7842 Name: Guillaume Insley
DateOfVisit: 2023-02-06
Symptoms: Hemorrhage, unspecified
Diagnosis: 4590
```

36. To **view** visit **by id** type 5 then enter the data for the fields that show.

```
Enter Visit ID or 0 to cancel: visit124

VisitID: visit124
Patient Info: ID: FZJXKBND Name: Skyler Chichgar
Doctor Info: ID: 9947 Name: Ole Hadcock
DateOfVisit: 2023-08-08
Symptoms: headache
Diagnosis: Animal drawn vehicle accident injuring other specified person
```

37. To **view** visits **by patients name and doctors name**type 6 then enter the data for the fields that show.

```
Enter patients first name or 0 to cancel: Janelle
Enter patients surname or 0 to cancel: Behnecken
Enter doctors first name or 0 to cancel: Ole
Enter doctors surname or 0 to cancel: Hadcock

VisitID: 1
Patient Info: ID: PHXNAQJ0 Name: Janelle Behnecken
Doctor Info: ID: 9947 Name: Ole Hadcock
DateOfVisit: 2023-05-06
Symptoms: Family history of other specified malignant neoplasm
Diagnosis: 168
```

Manage Conflicts

38. To **create** a conflict type 1 then enter the data for the fields that show.

```
Enter Drug ID or 0 to cancel: 55269
Enter ID of conflicting Drug or 0 to finish: 99089
```

39. To **delete** a conflict type 2 then enter the data for the fields that show.

```
Enter Drug ID or 0 to cancel: 55269
Enter ID of conflicting Drug or 0 to finish: 88506
```

40. To **view** conflicts **by drug id** type 3 then enter the data for the fields that show.

```
Enter Drug ID or 0 to cancel: 55269
Conflicting drugs
ACTIVATED CHARCOAL (88506)
SODIUM BICARBONATE (99089)
```

Testing

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Testing

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Testing Plan

Test Strategy and Methodologies

- **Unit Testing:** Testing individual methods (e.g., validating that a "Patient" object is created correctly).
- **Integration Testing:** Ensuring the Java application communicates correctly with the Hospital database to save and retrieve data.
- **System Testing:** Testing the complete application from the perspective of a user following the User Guide.
- **Error Handling (Black Box Testing):** Verifying that the program appropriately handles invalid inputs with exceptions or pre-validation (e.g., preventing string input in numeric field)

Test Plan Structure

- **Reliability Testing:** Scenarios to ensure the system doesn't crash. For example, testing what happens if a user enters a string when an integer ID is required.
- **Performance Testing:** Assessing how the system handles the "hard limit" on records shown (e.g., viewing 100+ visits) to ensure it remains responsive.
- **Database Integrity:** Verifying that creating, updating, and deleting records in Java correctly reflects in the database.

Test Types

White Box Testing (Structural Testing)

This method focuses on the internal code logic, branches, and loops rather than just the user inputs.

- **Unit Testing:** Testing individual methods within the classes (e.g., verifying a specific calculation in the Prescription class) to ensure they work in isolation.
- **Statement Coverage:** Ensuring every line of the Java code is executed at least once during the testing process to find hidden bugs.
- **Path Testing:** Testing all possible paths through a switch or if-else block, such as the different branches in the navigation menus.

Gray Box Testing

This is a combination of both Black and White box techniques. Use knowledge of the database structure to test the application's functionality.

- **Integration Testing:** Verifying that the Java application correctly interacts with the Hospital database. Look at the database tables to see if a record added in the app actually appeared in the SQL table.
- **Regression Testing:** Re-running previous tests after a bug is fixed (like the InputMismatchException mentioned in the test log) to ensure the fix didn't break other parts of the system.

Non-Functional Testing

The marking scheme explicitly requires coverage of **Reliability and Performance**.

- **Reliability Testing:** Stressing the system by entering invalid data types or extreme values to ensure the "program appropriately handles errors with exceptions" without crashing.
- **Performance (Load) Testing:** Testing how the system handles the "hard limit" on records shown. For example, might test how quickly the app displays 50 visits versus 5 visits to evaluate data retrieval speed.

Test Log

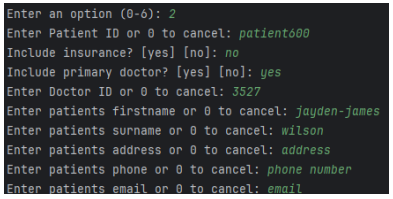
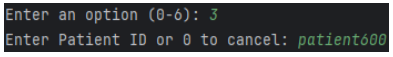
Test case	Test data	Expected result	Actual result	Comments	State

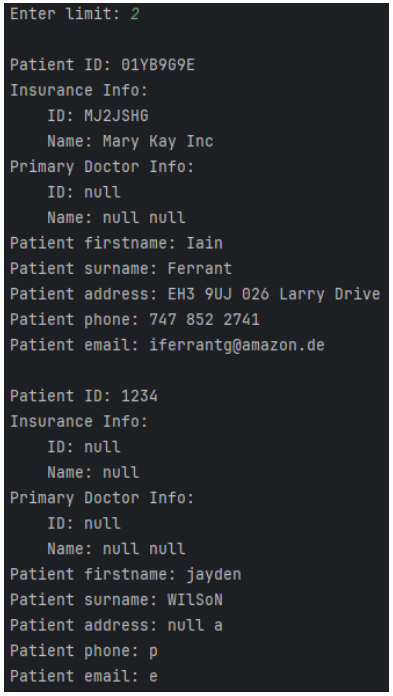
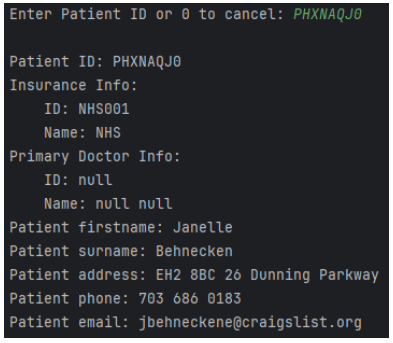
Early cancellation (global)	Patient ID: 0
	Insurance ID: 0
	Doctor ID: 0
	Doctors firstname: 0
	Insurance company name: 0
	patients address: 0
	Drug name: 0
		When 0 is entered the system should cancel the operation early	When 0 is entered the system cancel the operation early	`Enter Doctor ID or 0 to cancel: 0`
				`Operation cancelled`
				Same for others
					pass

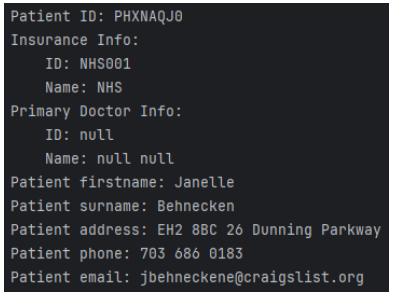
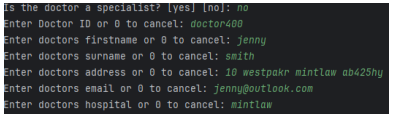
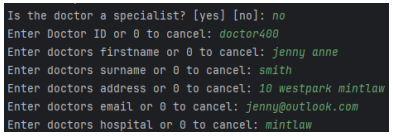
Global Negative Value Handling (All ID and fields across Doctors, Patients, Insurance, etc.)	-1, @#$%, NULL	System displays a "Validation Error" or "Invalid Input" message; system does NOT crash or save the record.	System displays error	`Enter Patient ID or 0 to cancel: -1` `Invalid input.`	pass

Date formatting	YYYY-MM-DD	System should only allow dates with the correct yyyy-mm-dd format	System only allows dates with the correct yyyy-mm-dd format	`Enter prescription date (yyyy-mm-dd) or 0 to exit: date`
				`Invalid date format. Please try again using YYYY-MM-DD.`
				`Enter prescription date (yyyy-mm-dd) or 0 to exit: 2025-04-90`
				`Invalid date format. Please try again using YYYY-MM-DD.`
					pass

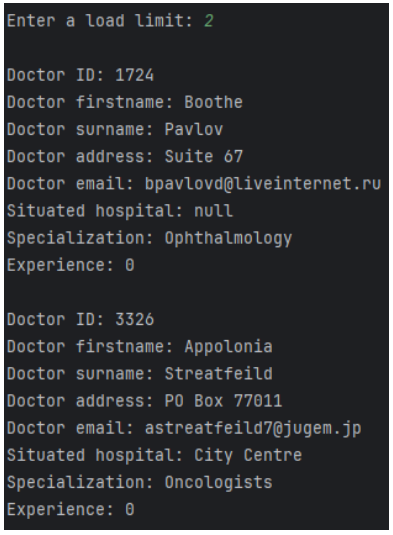
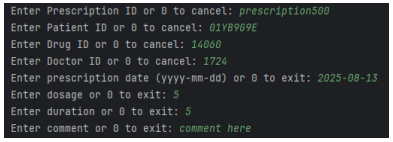
Performance (Retrieve 100 records using "Limit) for all modules	Limit: 100	Display within < 2 seconds	Display within < 2 seconds		pass
(Yes/No) Inputs (valid)	Include insurance?: no, n Include doctor?: no, n	When “no” is entered it should skip asking for that specific detail	Does in fact skip when no, n are entered	Include insurance? [yes] [no]: *no* Include primary doctor? [yes] [no]: *no*	pass
(Yes/No) Inputs (invalid)	Include insurance?: maybe Include doctor?: maybe	When “maybe” is entered it should ask for input again	Does ask for input again	Include insurance? [yes] [no]: *maybe* Invalid option. Please try again.	pass
Identifiers (Global)	Doctor Id: 0765 Insurance Id: 234567 Patient Id: 34567 Drug Id: 7129aba82 Visit id: 928272 Prescription id: 98765	When an id is entered that doesn't match an existing record it asks for id again unless “0” is entered for early cancellation	Asks for an id until a corresponding record is found	Enter Insurance ID or 0 to cancel: *234567* Insurance not found. Try again. Works the same for the rest	pass
Add Patient	Patient ID: patient600 Include insurance? [yes] [no]: yes Insurance ID: MJ2JSHG Include primary doctor? [yes] [no]: yes Doctor ID: 3527	Should add patent to database	Adds patient to database	Enter an option (0-6): *1* Enter Patient ID or 0 to cancel: *patient600* Include insurance? [yes] [no]: *yes* Enter Insurance ID or 0 to cancel: *MJ2JSHG* Include primary doctor? [yes] [no]: *yes* Enter Doctor ID or 0 to cancel: *3527* Enter patients firstname or 0 to cancel: *jayden* Enter patients surname or 0 to cancel: *wilson* Enter patients address or 0 to cancel: *10 westpark mintlaw, AB425HY* Enter patients phone or 0 to cancel: *07368350709* Enter patients email or 0 to cancel: *jayden@gmail.com*	pass

	patients firstname: jayden patients surname: wilson patients address: 10 westpark mintlaw, AB425HY patients phone: 07368350709 patients email: jayden@gmail.com				
Update patient	Patient ID: patient600 Include insurance? [yes] [no]: no Include primary doctor? [yes] [no]: yes Doctor ID: 3527 patients firstname: jayden-james Patients surname: wilson patients address: address patients phone: phone number patients email: email	Should update patient	Does update patient	 Email should check for “@” symbol but does not impact functionality	pass
Delete patient	Patient id: patient600	Should delete patient	Does delete patient		pass

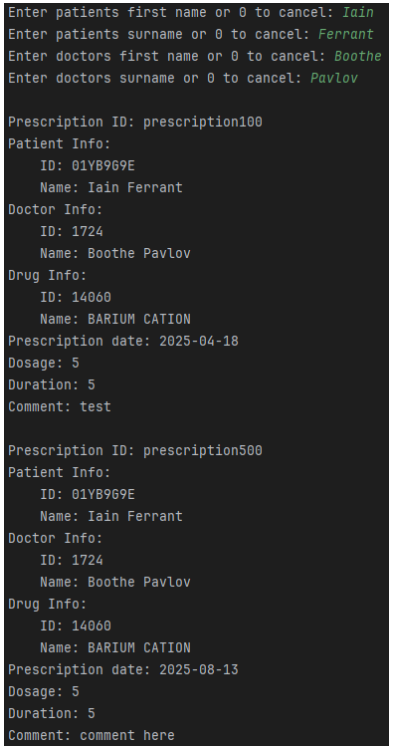
View all patients	Limit: 2	Should show all patient information within the limit	Does show all patient details within the limit		pass
View patient by id	Patientid: PHXNAQJ0	Should show details of patient by id	Does show details of patient by id		pass

View patient by first name and surname	Patient firstname: Janelle Patient surname: Behnecken	Shows a list of patients that match the same first name and surname	Shows a list of patients that match the same first name and surname		pass
Add doctor	Is the doctor a specialist? [yes] [no]: no DoctorId: doctor400 doctors firstname: jenny doctors surname: smith doctors address: 10 westpakr mintlaw ab425hy doctors email: jenny@outlook.com doctors hospital: mintlaw	Adds a new doctor	Adds a new doctor		pass
Update doctor	Is the doctor a specialist? [yes] [no]: no Doctor Id: doctor400 doctors firstname: jennyanne doctors surname: smith doctors address: 10 westpark mintlaw doctors email: jenny@outlook.com doctors hospital: mintlaw	Updates doctor	Updates doctor		pass

Delete doctor	doctorId: doctor400	Deletes doctor	Deletes doctor	Enter an option (0-7): 3 Enter Doctor ID or 0 to cancel: doctor400	pass
View doctor by id	doctorId: 3326	View doctor details by id	View doctor details by id	Enter Doctor ID or 0 to cancel: 3326 Doctor ID: 3326 Doctor firstname: Appolonia Doctor surname: Streatfeild Doctor address: PO Box 77011 Doctor email: astreatfeild7@jugem.jp Situated hospital: City Centre Specialization: Oncologists Experience: 0	pass
View doctor by first name and surname	doctors firstname: jenny doctors surname: smith	View doctor details by first name and surname	View doctor details by first name and surname	Enter doctors firstname or 0 to cancel: jenny Enter doctors surname or 0 to cancel: smith Doctor ID: doctor400 Doctor firstname: jenny Doctor surname: smith Doctor address: 10 westpakr mintlaw ab425hy Doctor email: jenny@outlook.com Situated hospital: mintlaw	pass
View doctors by speciality	speciality: 2 (NEUROLOGY)	View doctors by speciality	View doctors by speciality	Specialization Options 1: CARDIOLOGY 2: NEUROLOGY 3: PEDIATRICS 4: ONCOLOGY 5: ORTHOPEDICS 6: PSYCHIATRY Enter doctors specialization (1-6) or 0 to cancel: 2 Doctor ID: doc10 Doctor firstname: jayden Doctor surname: wilson Doctor address: address Doctor email: email Situated hospital: hospital Specialization: NEUROLOGY Experience: 10	pass

View all doctors	Limit: 2	View details of all doctors	View details of all doctors		pass
Add prescription	Prescription ID: prescription100 Patient ID: 01YB9G9E Drug ID: 14060 Doctor ID: 1724 prescription date (yyyy-mm-dd) : 2025-04-18 dosage: 5 duration: 3 comment: test	Add a prescription	Fails to add prescription as drug id does not exist. Cause: Constructor parameter order when creating a new prescription was in the wrong order. But is now fixed		Fail

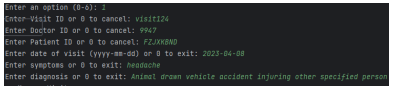
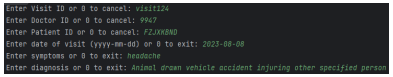
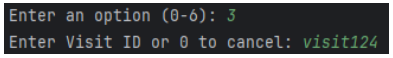
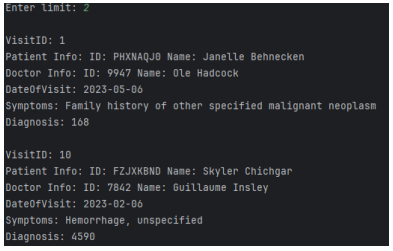
Update prescription	Prescription ID: prescription100 Patient ID: 01YB9G9E Drug ID: 15493 Doctor ID: 1724 prescription date (yyyy-mm-dd) : 2025-04-18 dosage: 5 duration: 3 comment: test	Update prescription	Update prescription	Enter an option (0-6): 2 Enter Prescription ID or 0 to cancel: prescription100 Enter Patient ID or 0 to cancel: 01YB9G9E Enter Drug ID or 0 to cancel: 1026382 Conflict found: BARIUM CATION (14060) Enter Drug ID or 0 to cancel: 15493 Enter Doctor ID or 0 to cancel: 1724 Enter prescription date (yyyy-mm-dd) or 0 to exit: 2025-04-04 Enter dosage or 0 to exit: 5 Enter duration or 0 to exit: 5 Enter comment or 0 to exit: 5 pills a week for 5 weeks	pass
Delete prescription	Prescription ID: prescription100	Delete prescription	Delete prescription	Enter an option (0-6): 3 Enter Prescription ID or 0 to cancel: prescription100	pass
View prescription by id	Prescription ID: prescription100	View prescription details by id	View prescription details by id	Enter Prescription ID or 0 to cancel: prescription100 Prescription ID: prescription100 Patient Info: ID: 01YB9G9E Name: Iain Ferrant Doctor Info: ID: 1724 Name: Boothe Pavlov Drug Info: ID: 14060 Name: BARIUM CATION Prescription date: 2025-04-18 Dosage: 5 Duration: 5 Comment: test	pass

View prescription by doctor and patient names	patients first name: Iain patients surname: Ferrant doctors first name: Boothe doctors surname: Pavlov	Display prescription details by patient and doctors names	Display prescription details by patient and doctors names		pass

View all prescriptions	Limit: 2	View details of all prescriptions within limits	View details of all prescriptions within limits	Enter limit: 2 Prescription ID: 1383892245 Patient Info: ID: FZJXKBND Name: Skyler Chichgar Doctor Info: ID: 9947 Name: Ole Hadcock Drug Info: ID: 34154 Name: risperidone Prescription date: 2023-04-08 Dosage: 16 Duration: 98 Comment: Animal-drawn vehicle accident injuring other specified person Prescription ID: 1634654226 Patient Info: ID: WSA59Z49 Name: Clevie Billsberry Doctor Info: ID: 4449 Name: Romola Thurber Drug Info: ID: 23810 Name: Aspen Prescription date: 2023-08-02 Dosage: 24 Duration: 34 Comment: Accidental poisoning by arsenic and its compounds and fumes	pass
Conflict during prescription management	Drug ID: 1026382	Display conflicts if found	Display conflicts if found. Conflict found: BARIUM CATION (14060)	Enter an option (0-6): 1 Enter Prescription ID or 0 to cancel: pre150 Enter Patient ID or 0 to cancel: 01YB9G9E Enter Drug ID or 0 to cancel: 1026382 Conflict found: BARIUM CATION (14060) Enter Drug ID or 0 to cancel:	pass
Add drug	Drug ID: drug315 drug name: moscovium side effects: death benefits: none to be found	Add drug	Add drug	Enter Drug ID or 0 to cancel: drug315 Enter drug name or 0 to exit: moscovium Enter side effects or 0 to cancel: death Enter benefits or 0 to cancel: none to be found	pass
Update drug	Drug ID: drug315 drug name: moscovium side effects: death benefits: none	Update drug	Update drug	Enter an option (0-6): 2 Enter Drug ID or 0 to cancel: drug315 Enter drug name or 0 to exit: moscovium Enter side effects or 0 to cancel: death Enter benefits or 0 to cancel: none	pass
Delete drug that is part of a prescription	Drugid: 14060	As the drug is part of existing prescriptions	As the drug is part of existing prescriptions its	Enter an option (0-6): 3 Enter Drug ID or 0 to cancel: 14060 (conn=247) Cannot delete or update a parent row: a foreign key constraint fails Need to add proper errors for	pass

		its prevented from being deleted	prevented from being deleted	foreign key errors	
View drug by id	Drugid: drug315	View details of drug by id	View details of drug by id	Enter Drug ID or 0 to cancel: drug315 Drug id: drug315 Drug name: moscovium Side effects: death Benefits: none	pass
View all drugs	limit : 2	View details of drugs within limit	View details of drugs within limit	Enter limit: 2 Drug id: 1026382 Drug name: test Side effects: test Benefits: test Drug id: 10987654 Drug name: test Side effects: test Benefits: test	pass
View Drug by drug name	Drug name: moscovium	View details of drug by drug name	View details of drug by drug name	Enter drugs name or 0 to cancel: moscovium Drug id: drug315 Drug name: moscovium Side effects: death Benefits: none	pass
Delete drug	Drugid: drug315	Delete drug	Delete drug	Enter an option (0-6): 3 Enter Drug ID or 0 to cancel: drug315	pass
Add insurance	Insurance Id: insurance145 insurance company name: HealthCo insurance address: 42 north	Add insurance	Add insurance	Enter Insurance ID or 0 to cancel: insurance145 Enter insurance company name or 0 to cancel: HealthCo Enter insurance address or 0 to cancel: 42 north avenue abereen AB436HW Enter insurance phone or 0 to cancel: 345678765	pass

	avenue abereen AB436HW insurance phone: 345678765				
Update insurance	Insurance Id: insurance145 insurance company name: HealthCo insurance address: 42 north avenue aberdeen AB436HW insurance phone: 345678765	Update insurance	Update insurance	Enter Insurance ID or 0 to cancel: insurance145 Enter insurance company name or 0 to cancel: HealthCo Enter insurance address or 0 to cancel: 42 north avenue aberdeen AB436HW Enter insurance phone or 0 to cancel: 345678765	pass
Delete insurance	Insurance Id: insurance145	Delete insurance	Delete insurance	Enter an option (0-6): 3 Enter Insurance ID or 0 to cancel: insurance145	pass
View all insurance	Limit: 3	View details of insurance within limit	View details of insurance within limit	Enter load limit: 3 Insurance ID: 54ARPD9 Company name: Cantrell Drug Company Address: 17668 Nova Pass Phone: 964-658-4951 Insurance ID: 9BFYJXV Company name: Carilion Materials Management Address: 068 Hollow Ridge Parkway Phone: 478-698-2522 Insurance ID: BS03EM0 Company name: Bay State Medical Address: 61735 Northfield Circle Phone: 212-489-0603	pass
View insurance by id	Insurance Id: insurance145	View details of insurance by id	View details of insurance by id	Enter Insurance ID or 0 to cancel: insurance145 Insurance ID: insurance145 Company name: HealthCo Address: 42 north avenue aberdeen AB436HW Phone: 345678765	pass
View insurance by company name	insurance company name: healthCo	View details of insurance by company name	View details of insurance by company name	Enter insurance company name or 0 to cancel: healthCo Insurance ID: insurance145 Company name: HealthCo Address: 42 north avenue aberdeen AB436HW Phone: 345678765	pass

Add visit	Visit ID: visit124 Doctor Id: 9947 Patient ID: FZJXKBND date of visit (yyyy-mm-dd): 2023-04-08 symptoms: headache diagnosis: Animal drawn vehicle accident injuring other specified person	Add visit	Add visit		pass
Update visit	Visit ID: visit124 Doctor Id: 9947 Patient ID: FZJXKBND date of visit (yyyy-mm-dd): 2023-08-08 symptoms: headache diagnosis: Animal drawn vehicle accident injuring other specified person	Update visit	Update visit		pass
Delete visit	Visit ID: visit124	Delete visit	Delete visit		pass
View all visits	Limit: 2	View details of visits within limit	View details of visits within limit		pass

View visits by id	Visit ID: visit124	View details of visit by id	View details of visit by id	Enter Visit ID or 0 to cancel: visit124 VisitID: visit124 Patient Info: ID: FZJXKBND Name: Skyler Chichgar Doctor Info: ID: 9947 Name: Ole Hadcock DateOfVisit: 2023-08-08 Symptoms: headache Diagnosis: Animal drawn vehicle accident injuring other specified person	pass
View visits by patient and doctors name	patients first name: Skyler patients surname: Chichgar doctors first name: Ole doctors surname: Hadcock	View visit details by patient and doctors name	View visit details by patient and doctors name	VisitID: visit124 Patient Info: ID: FZJXKBND Name: Skyler Chichgar Doctor Info: ID: 9947 Name: Ole Hadcock DateOfVisit: 2023-08-08 Symptoms: headache Diagnosis: Animal drawn vehicle accident injuring other specified person	pass
Add conflict	Drug ID: 55269 ID of conflicting Drug: 99089	Add conflict	Add conflict	Enter Drug ID or 0 to cancel: 55269 Enter ID of conflicting Drug or 0 to finish: 99089	pass
View Conflicts by Drug id	Drug ID: 55269	View conflicts details by Drug id	View conflicts details by Drug id	Enter Drug ID or 0 to cancel: 55269 Conflicting drugs ACTIVATED CHARCOAL (88506) SODIUM BICARBONATE (99089)	pass
Delete conflict	Drug I: 55269 ID of conflicting: 88506	Delete conflict	Delete conflict	Enter Drug ID or 0 to cancel: 55269 Enter ID of conflicting Drug or 0 to finish: 88506	pass

Evaluation

Software Development: Hospital Project

Evaluation

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Fulfillment of Specification

The implemented solution successfully meets the requirements outlined in the project brief.

- **Functionality**: The application provides a working Command Line Interface (CLI) that allows for the creation, retrieval, updating, and deletion (CRUD) of records for Patients, Doctors, Visits, and Prescriptions. I also implemented additional drug conflict management to ensure patient safety.
- **Database Connectivity:** The system is successfully integrated with the Hospital database using Java Database Connectivity (JDBC), ensuring data persistence across sessions.
- **Object-Oriented Design:** The code adheres to the required design principles, including the use of classes, constructor methods, and private class-wide variables to prevent content coupling.
- **Error Handling:** The application appropriately handles common user errors, such as invalid data types and database connection failures, through pre-validation and exception handling.

Conduct of the Software Development Project

The project followed a standard software development lifecycle (SDLC) consisting of design, implementation, and testing.

- **Design Phase**: Creating the UML diagrams (Class and Use Case) was critical for mapping out the relationships between entities.
- **Implementation Considerations:**
    - **Separation of Concerns:** I created a utility function to combine strings (e.g., Firstname and Surname) within the business logic rather than relying on SQL concat, adhering to better programming practices. Although some code still using concat in the sql queries due to time constraints.
    - **Data Structures:** I utilized a **HashSet** for loading conflicting drugs to prevent duplicates and ensure faster lookups.
    - **Refinement:** I separated "Conflicts" into their own class to improve error handling and debugging, even though they were originally planned as part of the Drug class.
- **Testing Phase:** I implemented a multi-layered testing strategy including Black Box, White Box, and Gray Box testing to ensure both the user interface and the back-end logic were robust.

Future Developments

Due to the strict deadlines of the assessment, certain features were prioritized over others. If more time were available, the following improvements would be implemented to enhance the "working solution":

- **Advanced Search Logic:** I would implement a feature allowing users to assign records (like Visits or Prescriptions) using names (e.g., "Doctor Smith") instead of just IDs. This would involve a background lookup to translate the name to an ID, making the system more user-friendly.
- **Expanded Business Logic:** While the string combination function was successful, time constraints meant it was only implemented in one module. I would expand this across the entire application for architectural consistency.
- **Graphical User Interface (GUI):** Moving from a CLI to a JavaFX or Swing GUI would improve accessibility and allow for better data visualization, such as trend tracking and modelling as suggested in the brief.

Knowledge and Skills Gained

Completing this project has significantly enhanced my technical proficiency:

- **Advanced Java Programming:** I gained a deeper understanding of inheritance and polymorphism by implementing specialized classes, such as `Specialist`, which extends the Doctor class. I also used an enum to hold types of specialisations
- **Data Integrity:** I learned to implement constraints in Java to prevent the deletion of records with active dependencies, protecting the reliability of the database.
- **Documentation:** Preparing a professional User Guide and a comprehensive Test Log improved my ability to communicate technical instructions to end-users.

Use of Feedback and Modifications

Feedback received during the formative stages led to several critical improvements:

- **User Experience (UX):** In response to feedback that "View All" lists were overwhelming, I implemented a "limit" variable to let users define the record count per view.
- **Input Validation**: Early testing showed crashes on invalid numeric inputs. I modified the code to include try-catch blocks for InputMismatchException and centralized this logic into a getInt() utility function for consistency.
- **Consistency**: I removed the standalone postcode field for patients, integrating it into the address field to match the format of other objects while maintaining backward compatibility so displaying patients still shows postcode using the corresponding database field.